

Appendix 1 – SMACnet Dypstick

Dypstick Overview [19]

The SMACnet Dypstick is a dynamic pointer consisting of two markers on a spring-loaded track (**Figure A1.1**). The linear displacement between *PointerTip* and *PointerTail* calculates a vector that drives the computation of a virtual marker at the location of the pointer's tip. **For individuals with excessive soft tissue over the location of the ASIS markers, the use of a Dypstick is strongly recommended to minimize soft tissue movement artefact.**

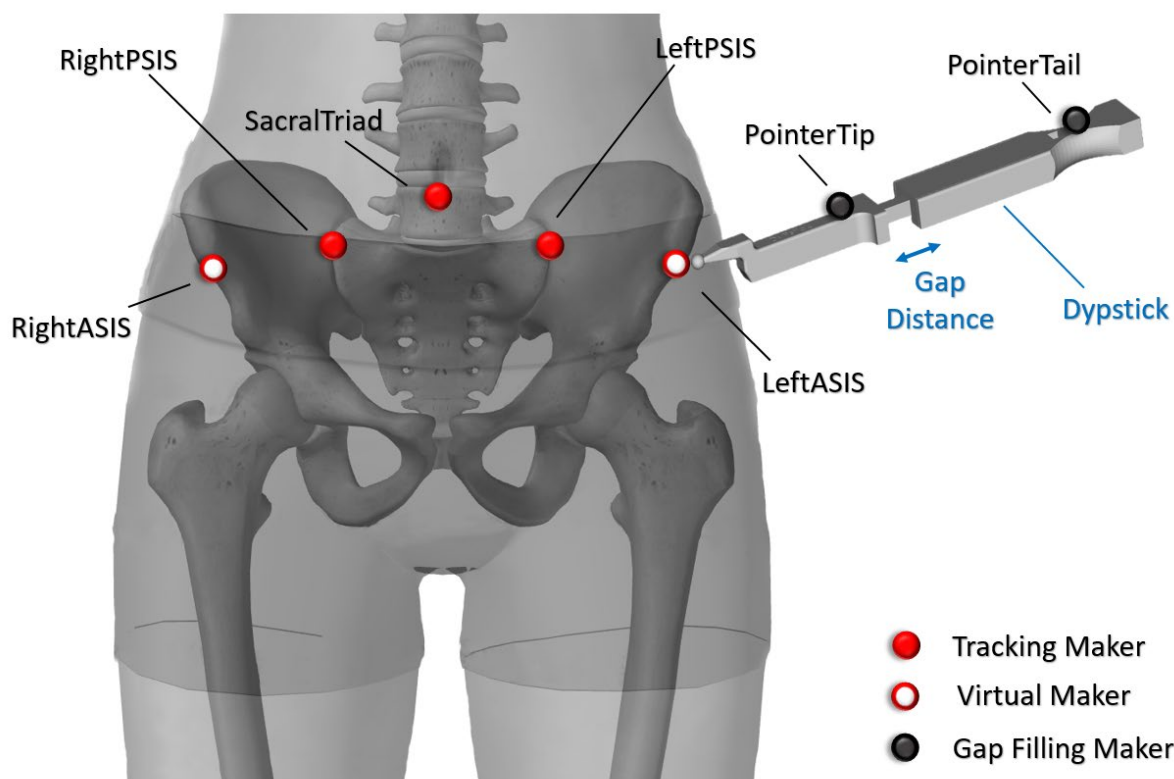


Figure A1.1: Schematic of the alternate pelvic marker set with example Dypstick use to define the ASIS markers virtually.

If a Dypstick is used to define virtual ASIS locations, an alternate marker set with three pelvic tracking markers is needed during data acquisition (**Table A1.1**). During the static standing trial, the pointer tip is placed at the location of the palpated landmark. The user depresses the device to define a vector from *PointerTip* to *PointerTail*. In post-processing, the virtual landmark is placed a known distance away from *PointerTip* along the direction of this vector. Known relationships between the virtually-defined ASIS locations and the pelvic tracking markers during the static calibration allow for the computation of virtual ASIS markers during walking trials.

Table A1.1: Alternate marker set for use in conjunction with the Dypstick for virtual ASIS definition. New markers are highlighted in yellow for clarity.

Marker Name	Landmark	Location
C7 ^{A,T}	7 th Cervical Vertebra	Prominence of C7 vertebra
Left_Clavicle, Right_Clavicle ^{A,T}	Sterno-clavicular	Thumb width below clavicle, equidistant from centerline of sternum
Sacral_Triad^{A,T}	Simulated sacrum	Superior to and equidistant from L/R PSIS
Left_P SIS, Right_P SIS ^{A,T}	Posterior superior iliac spine	Directly on left/right posterior superior iliac spine. Look for the dimples.
Left_Patella, Right_Patella ^T	Patella	On superior pole of patella
Left_Lateral_Knee, Right_Lateral_Knee ^{A,T}	Knee Joint (lateral)	Posterior lateral condyles at center of a visualized circle
Left_Medial_Knee.Cal, Right_Medial_Knee.Cal ^A	Knee Joint (medial)	Posterior medial condyles at the center of a visualized circle, which could be fit to this anatomy
Left_Lower_Tibia, Right_Lower_Tibia ^T	Tibia	On the anterior crest of lower third of tibia
Left_Lateral_Malleolus, Right_Lateral_Malleolus ^{A,T}	Ankle (lateral malleolus)	Lateral malleoli so that marker center lies on ankle flexion/extension axis.
Left_Medial_Malleolus.Cal, Right_Medial_Malleolus.Cal ^A	Medial malleolus	Medial malleoli so marker center lies on ankle flexion/extension axis.
Left_Heel, Right_Heel ^A	Heel	Medial/lateral center of posterior calcaneus
Left_2nd3rd_MT_Head, Right_2nd3rd_MT_Head ^{A,T}	Toe	Dorsal aspect of foot between 2nd and 3rd metatarsal heads (above metatarsal-phalangeal joints)
Pointer_Tip.Cal/Pointer_Tail.Cal	Pointer markers Tip and Tail	Device markers a known distance away from the Dypstick pointer tip. Used to virtually define ASIS markers

NOTE: Critical alignment is omitted from this table, as marker critical alignment for all other markers is unchanged from §3.1.

Dypstick Components

The 3D Printed assembly contains five (5) parts (**Figure A1.2**). Parts are numbered below corresponding to the numbers given in the exploded view.

1. Tip
 - a. Features ball head for palpation
 - b. Proximal marker (*Pointer_Tip*) located 120mm from ball head center
2. Sliding Rod
 - a. Thicker end of the sliding rod is glued and press-fit in **Part 1**
 - i. NOTE: Follow orientation showed in figure above.
 - b. Sliding rod free-floats in **Part 3**, allowing 25 mm translation when assembled
3. Base
 - a. Base contour is shaped to provide a secure hold without obstructing the marker
 - b. Distal marker (*Pointer_Tail*) located 182 mm from *Pointer_Tip*.
4. Spring
 - a. Provides stiffness for the sliding of the rod
5. Cover
 - a. Cover clips to the base, opened with flathead screwdriver to replace **Part 4**

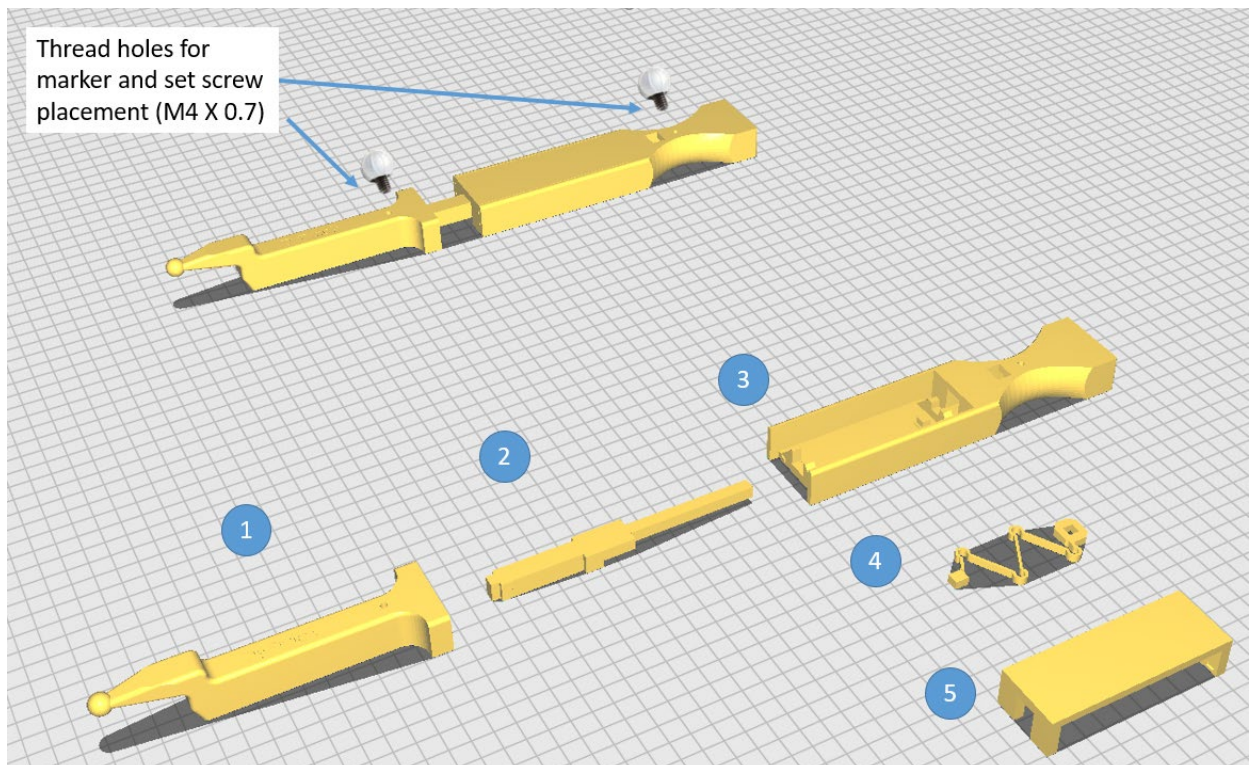


Figure A1.2: Dypstick assembly and exploded view

3D Print Specifications

1. 3D Printer minimal requirements
 - a. Layer resolution: 0.1 mm
 - b. Build Volume: 230L X 150W X 150H mm
2. 3D Printing specifications:
 - a. Material: 1.75 mm PLA
 - b. Nozzle temperature: 210 °C or as per PLA supplier's recommendations
 - c. Bed temperature: 60 °C or as per PLA supplier's recommendations
 - d. Configuration: "Fine" setting (layer height of 0.16 mm)
 - e. Material Infill: 20%
 - f. Supports:
 - i. Add light supports (line pattern preferable) for overhangs angle 45° and over
 - ii. Add box for nose head. Merge nose head's first slice with box's last slice for better results (**Figure A1.3**).

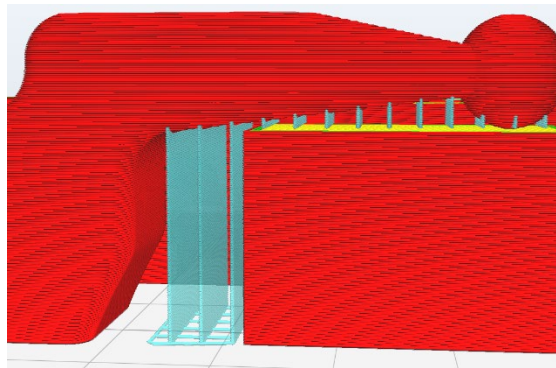


Figure A1.3: Line support and box support for nose head

- iii. Addition: No brim or raft necessary, skirt only
- iv. Printing orientation: **Figure A1.4**

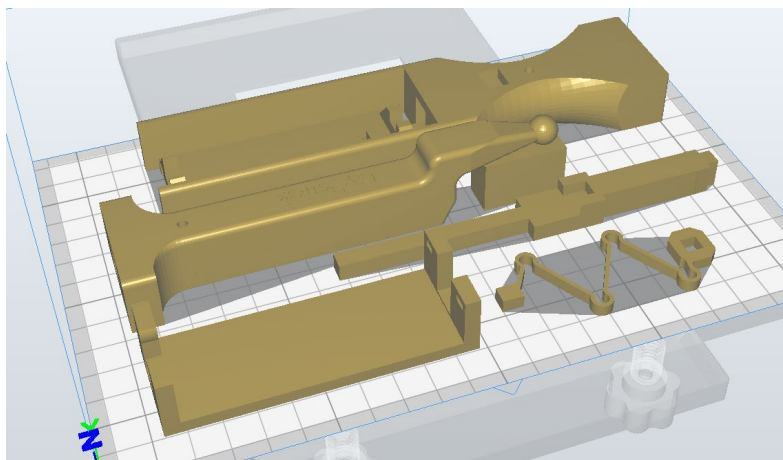


Figure A1.4: Print orientation on bed

Assembly and Validation

Material list, post-processing and assembly information have been provided below for a standard build of the Dypstick and relevant calibration tools.

Material List

- One set (1X) 3D-printed Dypstick assembly (**Figure A1.2**)
 - NOTE: Manual thread tapping of through-holes is required
- Two (2X) 9.5 mm pearl markers
- Two (2X) Set screw for fastening markers, M4 x 0.7 mm Thread, 10 mm Long

Validation

An annual validation of the Dypstick's accuracy is required. A Dypstick Calibration Plate with probing point is used to perform this test. The aim of the test is to reconstruct a virtual marker in the Dypstick Calibration Plate's probing point in different locations of the lab, while using the Dypstick in several orientations to ensure accuracy of the instrument (**Figure A1.5**). Steps are listed on the following page.

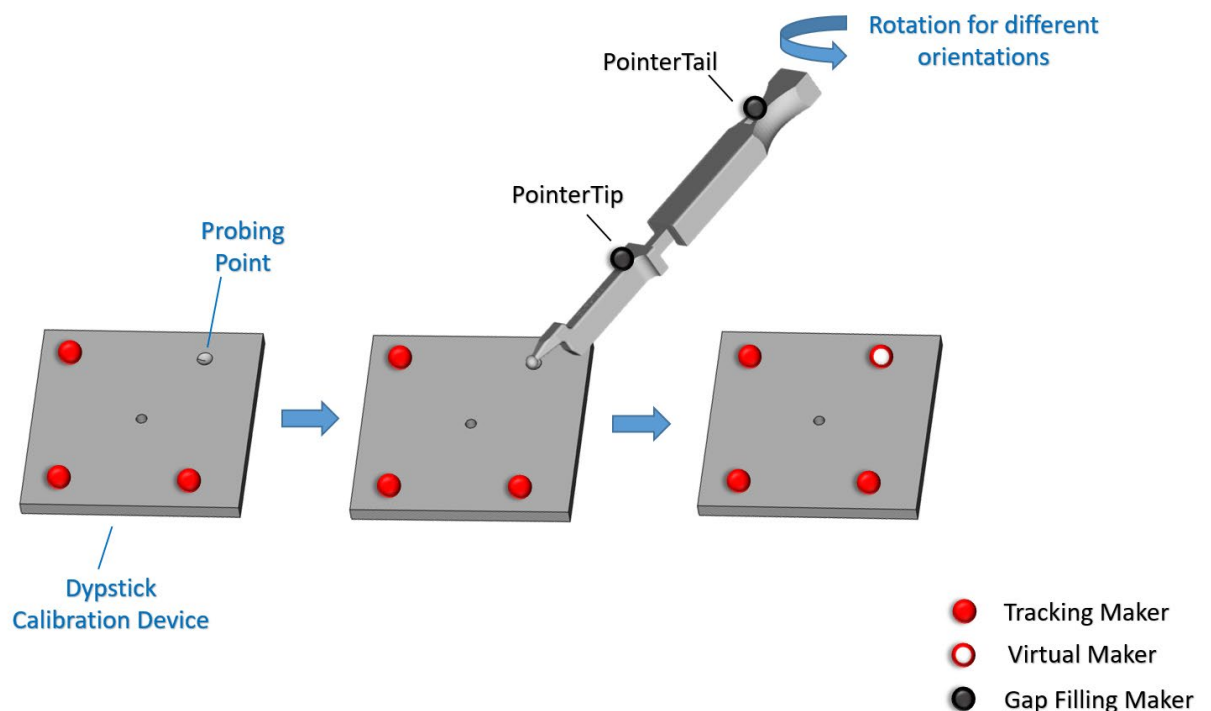


Figure A1.5: Dypstick validation

Validation Steps

1. Define five (5) locations in the lab that are in a typical acquisition perimeter
2. For each location:
 - a. Tape the base of the Calibration Plate on the ground
 - b. Perform an acquisition consisting of a Dypstick depression in the probing location of the Calibration Plate. Do this step three (3) times with the Dypstick in a different orientation.
3. When fifteen (15) acquisitions are recorded, export the data and:
 - a. Generate a coordinate system defined by the three (3) markers fixed on device (pick one of the three markers as origin).
 - b. Compute the virtual marker location in relation to the coordinate system generated.
 - c. Compare the virtual marker position for each trial.

The Dypstick print is considered validated if virtual marker positions are within 0.5 mm in relation to the origin of the Dypstick Calibration Plate's coordinate system. If it is not the case, re-print another Dypstick while using the 3D Print Specifications described above.

Dypstick Calibration Plate Construction

The Dypstick Calibration Plate presented below was developed specifically to ensure quality of data collection when using a Dypstick to define virtual markers. Material list, post-processing and assembly information are detailed below.

Material List

- One (1X) 3D-printed Dypstick Calibration Plate
- Three (3X) 9.5 mm pearl markers
- Three (3X) set screws for fastening markers
 - Thread: M4 x 0.7 mm
 - Length: 10 mm

NOTE: Manual thread tapping is required (through all) for screw holes

3D Print specifications

1. 3D Printer minimal requirements:
 - a. Layer resolution: 0.1 mm
 - b. Build Volume: 230L X 150W X 150H mm
2. 3D Printing specifications:
 - a. Material: 1.75 mm PLA
 - b. Nozzle temperature: 210 °C or as per PLA supplier's recommendations
 - c. Bed temperature: 60 °C or as per PLA supplier's recommendations
 - d. Configuration: "Fine" setting (layer height of 0.16 mm)
 - e. Material Infill: 20%
 - f. Supports: No supports needed
 - g. Addition: No brim or raft necessary, skirt only
 - h. Printing orientation (Error! Reference source not found.)

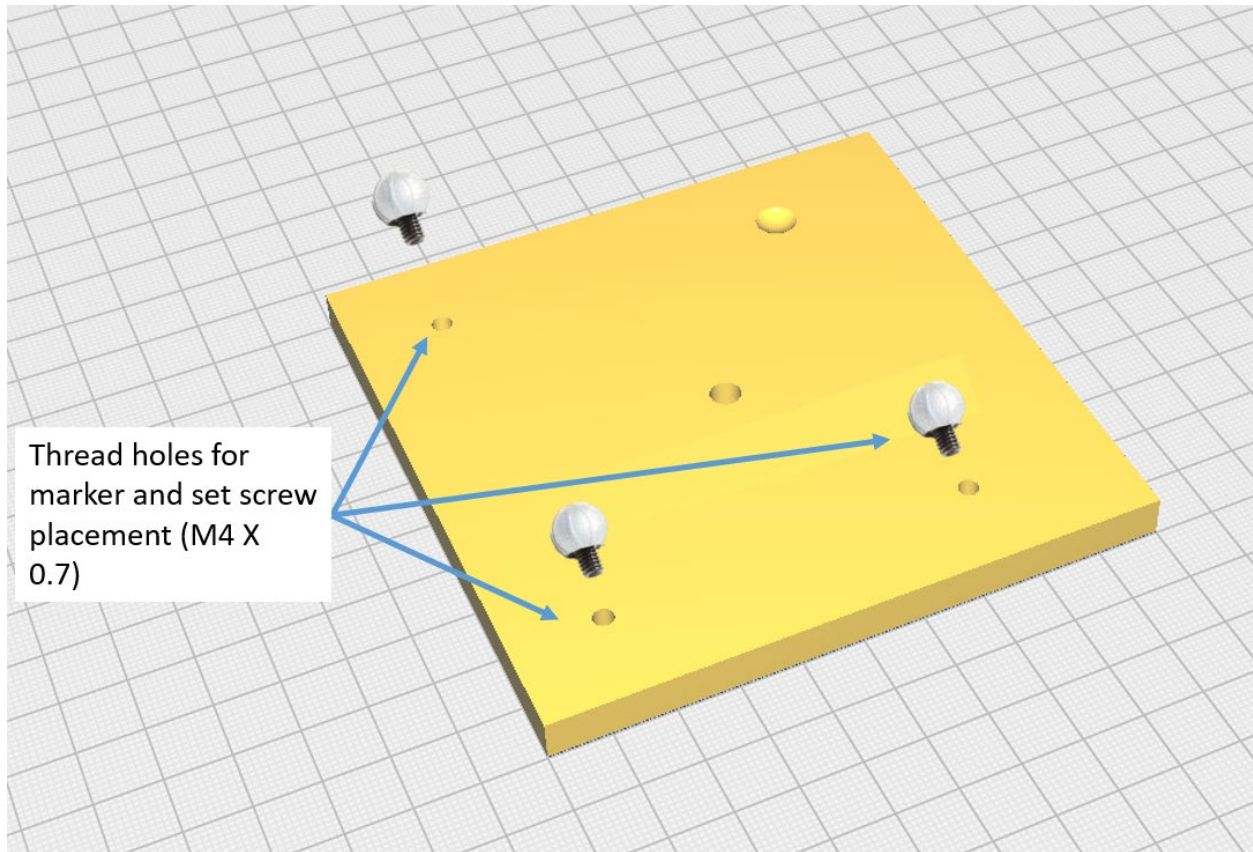


Figure A1.6 – Dypstick Calibration Plate Assembly